

To What Extent Do AgentFormer Attention Weights Give Faithful Explanations of the Interactions That Influence Pedestrian Trajectory Predictions?

DAMO 699 Capstone. Summary report. Draft for review.

1. Project Overview

Transformer models set the state of the art in pedestrian trajectory prediction. The models stay hard to interpret. Many teams look at an attention heat map and call the picture an explanation. Earlier work warns against that practice (Jain and Wallace, 2019; Wiegrefe and Pinter, 2019). This study replaces the visual read with a measurement. The study reads the AgentFormer attention map as a graph over the pedestrians of one prediction window. The study masks one edge into one ego pedestrian, runs the model a second time, and measures how far the forecast of that pedestrian moves. The study trains no model. The study uses the 5 public AgentFormer checkpoints at inference only.

2. Problem Statement

AgentFormer holds an explicit interaction graph. The attention map gives one weight for each ordered pair of agents. A user of the model wants to know whether that weight names the interaction that drives the forecast. The study answers one question. To what extent do the AgentFormer attention weights give faithful explanations of the interactions that influence the pedestrian trajectory predictions? The study measures faithfulness, and not causality. Every result describes the model. No result describes how a real person walks.

3. Objectives and Hypotheses

The study tests 3 primary hypotheses. Each null hypothesis states that the attention carries no extra information.

ID	Alternative hypothesis
H1	The removal of the strongest attention edge moves the forecast more than the removal of the weakest edge.
H2	The removal of the strongest attention edge moves the forecast more than the removal of the nearest neighbour edge.
H3	The context of a window, such as the crowd density, the agent count, the closing speed and the inverse time to collision, explains the faithfulness index of that window.

4. Dataset Description

The study uses the ETH and UCY benchmark. The benchmark holds 5 scenes of real pedestrians in world coordinates at 2.5 Hz. The data is public and anonymous. AgentFormer released its 5 checkpoints on the same benchmark, one checkpoint per held-out scene. The clean trajectory table holds 66,676 rows and 1,950 pedestrians. One window holds 8 observed frames, which is 3.2 s, and 12 predicted frames, which is 4.8 s. The frozen design keeps a window stride of 5 frames and an edge floor of 2 edges into the ego pedestrian. That design gives 481 eligible windows. The scene univ holds 60 percent of the rows, and the scene eth keeps only 8 windows.

5. Methodology

The pipeline runs 6 removal arms per window. The arms are MoRF, LeRF, nearest neighbour, random, a mass-matched control and a single-edge sweep. The primary case removes exactly 1 edge. The shift metric is the mean displacement between the unmasked forecast and the masked forecast of the ego pedestrian. The faithfulness index places the shift of the top attention edge on a known scale. The floor of that scale is the mean edge of the window, and the ceiling is the best edge. The full run takes 43,041 forward passes on 1 graphics card.

The design answers the independence concern of the proposal review. The windows overlap, the pedestrians repeat, and only 5 scenes exist. The cluster unit is therefore the connected component of the graph of ego pedestrians against time blocks, inside a scene. The 481 windows give 72 components. Each p value comes from a sign-flip permutation over those components, with 10,000 permutations. Each interval comes from a pairs cluster bootstrap, with 10,000 resamples. A test chooser picks the sign test when the differences are skew, and the Wilcoxon signed-rank test when the differences are symmetric. The Holm correction covers the 3 primary tests. The Benjamini-Hochberg correction covers the 7 supporting tests.

A gate protects the study. The attention score must reach an area under the curve above 0.9 on a synthetic scene with a planted graph before any real number counts.

6. Results

The pipeline passes every sanity check. Two identical calls give identical paths, so the zero check is 0.0 in all 5 scenes. AgentFormer is deterministic, so the noise floor is also 0.0. Any shift above 0 is therefore a real change of the forecast. The gate passes with an area under the curve of 0.9270, and the reversed control gives 0.0730. The pooled forecast error is 0.878 m average displacement and 1.982 m final displacement over the 481 windows.

ID	Effect	95 percent interval	Adjusted p	Verdict
H1	MoRF beats LeRF in 85.2 percent of 481 windows. The median shift gain is 0.039 m.	0.809 to 0.907	0.0003	Reject the null
H2	MoRF beats the nearest neighbour in 80.4 percent of the 347 non-zero windows. The median gain is 0.0087 m.	0.756 to 0.846	0.0003	Reject the null
H3	The 4 context terms explain 1.7 percent of the variance that the scene leaves. Partial R squared is 0.017.	not defined	0.60	Do not reject

The tests run over 72 components for H1 and H2, and over 62 components for H3. Three supporting tests strengthen H2. The Spearman correlation of the attention with the distance is only -0.23, so the attention is not a copy of the distance. The partial correlation of the attention with the shift, at a fixed distance, is 0.70. A shuffle of the attention weight inside a scene destroys the result, with a p of 0.0001. One supporting test limits the claim. A mass-matched set of light edges beats the single top edge, because MoRF wins in only 27.0 percent of those pairs.

The findings hold across the scenes. H1 points the same way in all 5 scenes, and the I squared value is 0.0 for both H1 and H2. H1 also holds at every removal step from 1 to 5 edges. H1 holds again under both other attention modules, under both other time collapses and under the second mask policy. H2 holds in univ, zara1 and zara2. H2 carries no information in eth, because attention and proximity name the same edge in 7 of the 8 eth windows.

7. Interpretation

The AgentFormer attention gives a faithful rank of the single edges that move its own forecast. The top edge beats the weakest edge and it beats the nearest neighbour, so proximity alone does not explain the rank. A reader must treat the map as a rank order, and not as a budget. The mass-matched control shows why. A sum of light weights does not carry the influence of one heavy weight, so the attention mass is not a currency of influence. The context of a window does not predict how faithful the attention is. A user therefore cannot switch trust on and off by the crowd density or by the closing speed. Every statement here describes the model and not the street.

8. Limits

- The released AgentFormer is deterministic. The 20 draws are 20 fixed modes and not 20 samples, so the study makes no probabilistic calibration claim.
- Only 5 scenes exist, and the scene univ holds 60 percent of the rows. The scene eth keeps 8 windows, so every eth number is weak.
- A window with exactly 2 edges gives only 2 reachable index values. The headline H3 fit therefore drops those 60 windows.
- The study measures faithfulness to the model. The study is not evidence about how people walk.

9. Recommendations and Next Steps

- Read an AgentFormer attention map as a rank of neighbours. Read the top edge, and ignore the size of the weight.
- Read the map beside the edge count of the window. A window with few neighbours carries almost no information.
- Add a count-controlled arm beside the mass-matched arm. The next design must vary the edge count and the edge mass one at a time.

- Repeat the same method on a stochastic forecaster, and then on another attention model. The cluster design, the permutation test and the bounded index need no AgentFormer.

10. Tools and Technologies

The pipeline runs on Python 3.12. AgentFormer runs on PyTorch with CUDA. NumPy and pandas build the 8 tables. SciPy, statsmodels and pingouin run the tests, the wild cluster bootstrap and the partial correlation. Matplotlib and seaborn draw the 12 figures. Pytest holds 422 tests, and all 422 pass. Git and GitHub hold the code.

11. Business Value and Real-World Impact

- A trust verdict. The study gives a statistics-backed answer on whether a team may read an attention map as evidence of a real interaction.
- A reusable audit method. The edge-ablation recipe, the bounded index and the proximity control transfer to any attention model that exposes an agent by agent map.
- A guardrail against a false explanation. The mass-matched result warns a team away from a dashboard that adds attention weights.

12. Deliverables

The repository holds the inference and ablation pipeline, 8 processed tables, 19 output tables, 12 figures, a read-only notebook and 422 tests. The file `docs/REPORT.md` holds the full method, every sensitivity table and every source path. The file `README.mmd` holds the pipeline diagram. This summary states the answer. It does not repeat the long report.